

Gravity Compensation Hydraulic Damper for Postural Transients

Executive Technology Brief

Domain: Hydraulic Valve Mechanics + Proportional Damping

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Postural ICP excursions cause over-drainage and under-drainage

Mechanism: Proportional hydraulic damper attenuates postural pressure transients continuously (vs ASD on/off switching)

Architecture: CSF shunt with hydraulic damper that proportionally attenuates postural pressure transients. Unlike fixed anti-siphon devices (ASD) that switch open/closed, this provides continuous proportional damping based on gravity head.

Governing Model: Hydraulic damper provides flow-proportional resistance. Postural transients excite the system; damper coefficient c_h determines settling time and overshoot.

Key Equations:

$dP_{gravity} = \rho * g * dh$ [hydrostatic pressure differential]

$F = c * v$ [linear viscous damper]

$dP_{total} = dP_{valve} + dP_{gravity} - dP_{damper}$

$dP_{damper} = c_h * Q$ [hydraulic damping]

Settling time: $\tau = I_h / c_h$ [first-order response]

Kill Condition: Damper coefficient cannot be tuned to both attenuate transients AND maintain minimum drainage floor

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.